

LECTURE 22 (NOVEMBER 26TH)

Proposition. (Feynman rules in the momentum space)

- (i) Construct all connected diagrams.
- (ii) For each arrowed leg, we have the assign the Dirac propagator

$$\left(\frac{-i}{\not{k} + m - i\epsilon} \right)_{\alpha\beta}$$

- (iii) For each wavy leg, in the Feynman gauge,

$$\frac{-i\eta_{\mu\nu}}{k^2 - i\epsilon}$$

- (iv) For each vertex,

$$ie(\gamma^\mu)_{\beta\alpha} \cdot (2\pi)^4 \delta^{(4)}(k_1 + k_2 - k_3)$$

- (v) Integrate over all the internal momenta.
- (vi) Divide by the symmtry factor.
- (vii) Consider the rearrangement of Grassmannian fields $\times (-1)^n$.
- (viii) Sum over all diagrams.

Example. ($e^-e^+ \rightarrow e^-e^+$ Bhabha scattering) The correlation function of our interest is:

$$G_{c\alpha_1\alpha_2;\beta_1\beta_2}^{(2;2;0)}(x_1, x_2, y_1, y_2) = \langle T\psi_{\alpha_1}(x_1)\psi_{\alpha_2}(x_2)\bar{\psi}_{\beta_1}(y_1)\bar{\psi}_{\beta_2}(y_2) \rangle$$

The Fourier transform is given as

$$\tilde{G}_{c\alpha_1\alpha_2;\beta_1\beta_2}^{(2;2;0)}(p_1, p_2; q_1, q_2)$$

Let's first compute in the position space! There are two possible diagrams.

$$\begin{aligned} & G_{c\alpha_1\alpha_2;\beta_1\beta_2}^{(2;2;0)}(x_1, x_2; y_1, y_2) \\ &= \int d^4z_1 d^4z_2 \left[(-1) \left(S_F(x_1 - z_1) ie\gamma^\mu S_F(z_1 - y_1) \right) D_{F\mu\nu}(z_1 - z_2) \left(S_F(x_2 - z_2) ie\gamma^\nu S_F(z_2 - y_2) \right) \right]_{\alpha_2\beta_2} \\ & \quad + \left(S_F(x_2 - z_1) ie\gamma^\mu S_F(z_1 - y_1) \right)_{\alpha_2\beta_1} D_{F\mu\nu}(z_1 - z_2) \left(S_F(x_1 - z_2) ie\gamma^\nu S_F(z_2 - y_2) \right)_{\alpha_1\beta_2} \end{aligned}$$

where $(m, n) = (1, 0)$. In the momentum space,

$$\begin{aligned}
& \tilde{G}_{c\alpha_1\alpha_2;\beta_1\beta_2}^{(2;2;0)}(-p', q; -p, q') \\
&= (-1) \int \frac{d^4 k}{(2\pi)^4} \left(\frac{1}{-\not{p} + m - i\epsilon} i e \gamma^\mu (2\pi)^4 \delta^{(4)}(p + p' - k) \frac{1}{\not{p} + m - i\epsilon} \right)_{\alpha_1, \beta_1} \\
& \quad \times \frac{-i\eta_{\mu\nu}}{k^2 - i\epsilon} \times \left(\frac{1}{\not{q} + m - i\epsilon} i e \gamma^\nu (2\pi)^4 \delta^{(4)}(k - q - q') \frac{1}{-\not{q}' + m - i\epsilon} \right)_{\alpha_2, \beta_2} \\
& \quad + \left[(-p', \alpha_1) \leftrightarrow (q_1, \alpha_2) \right] \\
&= \delta^{(4)}(p + p' - q - q') \left[(-1) \left(\cdot \right)_{\alpha_1 \alpha_2} \times \frac{-i\eta_{\mu\nu}}{(p + p')^2 - i\epsilon} \left(\cdot \right)_{\alpha_2 \beta_2} + \dots \right]
\end{aligned}$$

Definition. (Mandelstam variables for $2 \rightarrow 2$ scattering) We define

- (i) $s = -(p + p')^2$, the square of the total initial momentum.
- (ii) $t = -(p - q)^2$, the square of the difference of the initial and final momenta of the most similar particles.
- (iii) $u = -(p - q')^2$, the interchangeable variable with t by swapping final momenta.

Let's make some comments. Bhabha scattering consists of a s -channel and a t -channel. In the case of Moller scattering, there is a t -channel and s -channel.

Example. ($e^- \gamma \rightarrow e^- \gamma$ Compton scattering)

$$\tilde{G}_{c\alpha;\beta;\mu\nu}^{(1;1;2)}(P; Q; K_1; K_2) = \tilde{G}_{c\alpha;\beta;\mu\nu}^{(1;1;2)}(q; -p; -K, K') w$$

The above is equal to

$$\begin{aligned}
& (2\pi)^4 \delta^{(4)}(p + K - q - K') \times \\
& \left[\frac{-i\eta_{\mu\rho}}{K^2 - i\epsilon} \frac{-i\eta_{\sigma\nu}}{K'^2 - i\epsilon} \left(\frac{-i}{\not{q} + m - i\epsilon} i e \gamma^\rho \frac{-i}{(\cancel{p+K}) + m - i\epsilon} i e \gamma^\sigma \frac{-i}{\not{p} + m - i\epsilon} \right)_{\alpha\beta} \right] \\
& + \left((K, \mu) \leftrightarrow (-K', \nu) \right)
\end{aligned}$$